

Please ignore this message until our meeting tomorrow. It contains detailed instructions for the lab you will be working on, and I am sharing it here in advance so that you can easily access it during the session.

1- Create a shared package that has 2 typedef enums

```
typedef enum {SHIFT, ROTATE} mode_e;
```

```
typedef enum {RIGHT, LEFT} direction_e;
```

2- Create a sequence item

- > Do the essentials

- > Add rand modifier to the class variables (use above enums)

- > Add constraint to disable the reset most of the time using dist operator

- > add convert2string and convert2string_stimulus functions

3- Create a reset and main sequences

- > Do the essentials

- > Add body task in the sequences

- > add the seq item as parameter to the parent class

4- Create a sequencer

- > Do the essentials

- > add the seq item as parameter to the parent class

5- Adjust the driver class

- > add the seq item as parameter to the parent class

- > Remove the config object, build phase and connect phase

- > Add get_next_item and item_done in the run_phase

6- Create the monitor

- > Do the essentials

- > Create vif, analysis port, and seq item

- > In the run phase broadcast the seq_item

7- Create an agent

- > Do the essentials

- > create the driver, sequencer and monitor

- > connect the vif of the driver and monitor

- > connect the ports for the sqr and driver

- > connect the analysis port of the monitor and agent

8- Create the scoreboard

- > create the analysis export, analysis fifo
- > connect the analysis exports
- > In the run phase, get the seq item and check for the output value

9- Create the coverage collector

- > create the analysis export, analysis fifo
- > connect the analysis exports
- > create a covergroup with coverpoints on all inputs without specific bins defined
- > create the covergroup in the constructor
- > In the run phase, get the seq item and sample the covergroup

10- Adjust the env class

- > Remove the driver and create the agent, scoreboard and coverage collector
- > Connect agent analysis ports with the scoreboard and coverage collector analysis exports

11- Adjust the test class

- > open the sequences on the sequencers in the run phase